Christopher Johnstone

PhD candidate, Chemical Engineering

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Education

Sep 2019 – Ongoing
Massachusetts Institute of Technology, Cambridge, MA
PhD candidate in Chemical Engineering, minor in biological systems modeling.
- *Advisor*: Kate E. Galloway
- *Thesis*: Supercoiling as a regulator: the role of biophysical feedback in transcription and chromatin dynamics.

Aug 2015 – Jun 2019
California Institute of Technology, Pasadena, CA
BS in Chemical Engineering, Minor in Computer Science.
- *Advisor*: David A. Tirrell
- *Senior thesis*: Control of aggregated bacterial communities through engineered surface displayed proteins.

Experience

Dec 2019 - Ongoing
Galloway Lab, MIT – Cambridge, MA
PhD candidate
- Investigated supercoiling as a biophysical method to control genes. Performed both foundational modeling work predicting circuit behavior and experimental work demonstrating these predictions in a variety of cell types for the first time, including hiPSCs.
- As a founding graduate student, developed lab norms, best practices, data handling pipelines, internal tools, and internal NGS pipelines.
- Created and maintained lab Python packages, contributed to the Julia SciML ecosystem.

Fall 2021
Dept. of Chemical Engineering, MIT – Cambridge, MA
TA for Graduate Thermodynamics
- Developed homeworks, wrote course exams, prepared review sessions, and wrote lecture notes.

Dec 2016 – Jun 2019
Tirrell Lab, Caltech – Pasadena, MA
Research student, Robb and Eunice Rutledge SURF fellow (summer 2017)
- Demonstrated a synthetic aggregation circuit and quorum sensing circuit to manipulate and sense aggregation of communities of *E. coli* cells.
- Rationally designed and characterized a protein for photocontrollable bacterial aggregation.

2017-2019
Dept. of Chemical Engineering, Dept. of Computer Science, Caltech – Pasadena, CA
TA for Separation Processes, Intro to Chemical Engineering Computation, Computer Graphics, Intro to Programming Methods
- Developed novel examples, practice tests, and review sessions for weekly recitation sessions.

Jun 2018 – Aug 2018
Provivi – Santa Monica, CA
Metabolic engineering intern
- Identified and overexpressed bottleneck pathway genes, creating a yeast strain producing 30% more lipid dry mass.

Jun 2016 – Sep 2016
Pure Storage – Mountain View, CA
Software engineering intern
- Implemented a customer-facing feature to accurately account utilized storage space across clusters.

Publications

2025

1. **Johnstone, C. P.** *et al.* Gene Syntax Defines Supercoiling-Mediated Transcriptional Feedback. *bioRxiv* (Jan. 19, 2025). Pre-published.
2. Love, K. S., **Johnstone, C. P.**, ... & Galloway, K. E. Model-Guided Design of microRNA-based Gene Circuits Supports Precise Dosage of Transgenic Cargoes into Diverse Primary Cells. *Cell Systems* (June 18, 2025).
3. Peterman, E. L., ..., **Johnstone, C. P.**, ... & Galloway, K. E. High-Resolution Profiling Reveals Coupled Transcriptional and Translational Regulation of Transgenes. *Nucleic Acids Research* (June 6, 2025).

2023

4. Kozlowski, M. T., ..., **Johnstone, C. P.**, ..., Tirrell, D. A. & Ku, H. T. A Matrigel-Free Method for Culture of Pancreatic Endocrine-like Cells in Defined Protein-Based Hydrogels. *Frontiers in Bioengineering and Biotechnology* (Mar. 9, 2023).

2022

5. **Johnstone, C. P.** & Galloway, K. E. Supercoiling-Mediated Feedback Rapidly Couples and Tunes Transcription. *Cell Reports* (Oct. 18, 2022).

2021

6. **Johnstone, C. P.** & Galloway, K. E. Engineering Cellular Symphonies out of Transcriptional Noise. *Nature Reviews Molecular Cell Biology* (Mar. 15, 2021).
7. Kozlowski, M. T., Silverman, B. R., **Johnstone, C. P.** & Tirrell, D. A. Genetically Programmable Microbial Assembly. *ACS Synthetic Biology* (June 18, 2021).

2020

8. **Johnstone, C. P.**, Wang, N. B., Sevier, S. A. & Galloway, K. E. Understanding and Engineering Chromatin as a Dynamical System across Length and Timescales. *Cell Systems* (Nov. 18, 2020).

Talks

2025

1. Genome Architecture and Function (Poster). *Gene syntax defines supercoiling-mediated transcriptional feedback.* Sofia, Bulgaria.
2. Engineering Biology Club, Ginkgo Bioworks. *Gene syntax defines supercoiling-mediated transcriptional feedback.* Boston, MA.

2024

1. March APS. *Supercoiling-mediated feedback rapidly couples and tunes transcription in mammalian cells.* Minneapolis, MN.

2023

2. Boston Mammalian Synthetic Biology. *Supercoiling reversibly couples adjacent gene expression through biophysical feedback*. Boston, MA
3. Mammalian Synthetic Biology Workshop (Poster). *Supercoiling-mediated transcriptional dynamics impart rapid, tunable feedback.* San Jose, CA.

2022

4. AIChE Annual Meeting. *Modeling Supercoiling-Dependent Feedback As a Transcriptional Coordinator to Understand and Engineer Biological Circuits.* Phoenix, AZ.
5. Boston Mammalian Synthetic Biology. *Supercoiling-mediated feedback rapidly couples and tunes transcription.* Boston, MA.

2018

6. 255th ACS National Meeting. Poster. *Strategies for controlled bacterial assembly resulting in activation of a quorum-sensing circuit.* (with M. T. Kozlowski). New Orleans, LA.

Leadership

Mar 2020 – Ongoing | **Sidney Pacific Graduate Residence, MIT**
VP of Residential Life, Web Chair Lead
- Secured funding for the residence, and led a group of officers in development of safe social events to build community amid pandemic restrictions. Led the team that develops software for residents and building staff.

Sep 2020 – Sep 2021 | **GSCX, MIT**
- Coordinated and organized virtual social events for the Chemical Engineering department throughout Covid.

Apr 2016 – Apr 2019 | **Board of Control, Caltech**
Board member, Secretary (until Mar 2018), Chair
- Led investigations, coordinated with administrators and professors, and organized hearings for academic honor code violations.

Honors

Jun 2019 | **Frederic W. Hinrichs Jr. Memorial Award Recipient**
Award given to the two seniors who have made the greatest contribution to the student body and whose qualities of character, leadership, and responsibility have been outstanding.

May 2018 | **Donald S. Clark Award Recipient**
Award given to two engineering juniors who show academic excellence, leadership, and dedication to the Caltech community.

Mar 2018 | **Elected to Tau Beta Pi**

Skills

- **Experimental**: Tissue culture, iPSC culture, lentivirus production, molecular cloning, genomics (sci-ATAC-seq, Micro-C, CUT&Tag, NGS library prep), FACS and flow cytometry, protein production and purification. Basic mouse training.
- **Computational**: Biological circuit modeling, stochastic simulations, bioinformatic and NGS pipelines. Computer graphics and networking programming experience. Extensive experience in Python, Julia, C++, C. Experience in Go, Haskell, MATLAB, TypeScript & JS, PHP.